Reg. No									

St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001

CH305T – INORGANIC CHEMISTRY - II

Time : 3 hrs **Max Marks :75**

SECTION – A (10X1=10)

Answer **ALL** Questions

I. Choose the correct answer

1. Brown ring test is used to detect
   a. NO_3^- b. Cl^- c. I^- d. Br^-

2. Which of the following is aprotic solvent?
   a. CH_3COOH b. C_6H_6 c. HCN d. HCOOH

3. Property of catenation is strongest in
   a. C b. O c. N d. Si

4. Silicon carbide is used
   a. As a dehydrating agent b. As an abrasive
   c. As a solvent d. As a catalyst

5. Which one is used as anaesthetic
   a. NH_3 b. N_2O c. NO d. NO_2

6. Caro's acid is
   a. $H_2S_2O_5$ b. $H_2S_2O_8$ c. H_2SO_3 d. H_2SO_5

7. The strongest reducing agent among following is
   a. F^- b. Cl^- c. Br^- d. I^-

8. Which does not form oxy acids
   a. F_2 b. Cl_2 c. Br_2 d. I_2

9. Most abundant noble gas in air
   a. He b. Ne c. Ar d. Kr

10. XeF_6 on complete hydrolysis gives
    a. Xe b. XeO_3 c. XeO_2 d. XeO_4

SECTION – B (10X2=20)

Answer any **TEN** Questions

11. What is solubility product?

12. Define vander Waal's force.

13. What is amphoteric solvent? Give examples.

14. Define catenation.

15. Complete the following equation.
    $C + 2H_2SO_4 \longrightarrow ? + ? + ?$

16. Draw the structure of NH_3 and AsH_3.

17. Mention uses of hydrazine.

18. What is caro's acid?

19. Write the electronic configuration of Cl and I.

20. Complete the following equation.
    $KClO_3 + I_2 \longrightarrow ? + ?$

21. What is pseudohalogen?

22. List out the noble gases.

23. Define hybridization.

24. Write the differences between acid-base reactions and oxidation reduction reactions.

SECTION- C (5X7=35)

Answer any **FIVE** Questions

25. Write the confirmation test for nitrate and Phosphate.

26. Explain the liquid ammonia as a solvent.

27. Discuss the comparison of carbon and silicon.

28. Classify the carbides and explain with examples to each of them.

29. Explain the preparation and properties of H_3PO_4.

30. Discuss the preparation, properties and uses of hydroxylamine.

31. a) Draw the structures of BrF5 and IF5 and find the hybridization of central atoms. **(2+2)**
    b) What are pseudo halogens? Give examples. **(3)**

32. a) Discuss the hybridization and geometries of XeF_6 and $XeOF_4$. **(2+2)**
    b) Cite any one reason for placing noble gases in zero group. **(1)**
    c) Write the uses noble gases. **(2)**

SECTION- D (1X10=10)

Answer any **ONE** Question

33. a. What are non-aqueous solvents? How are they classified? Give an example each type. **(6)**
    b. Explain the applications of carbides in Industry. **(4)**

34. What are hydrides of nitrogen group? Write their structure, trends in boiling point, basic character and hydrogen bonding. **(10)**

35. a) Write the exceptional properties of fluorine. **(4)**
    b) Write the similarities between H_2O&HF. **(2)**
    c) Discuss the structures of sodium hypochloride and sodium chlorite. **(4)**

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